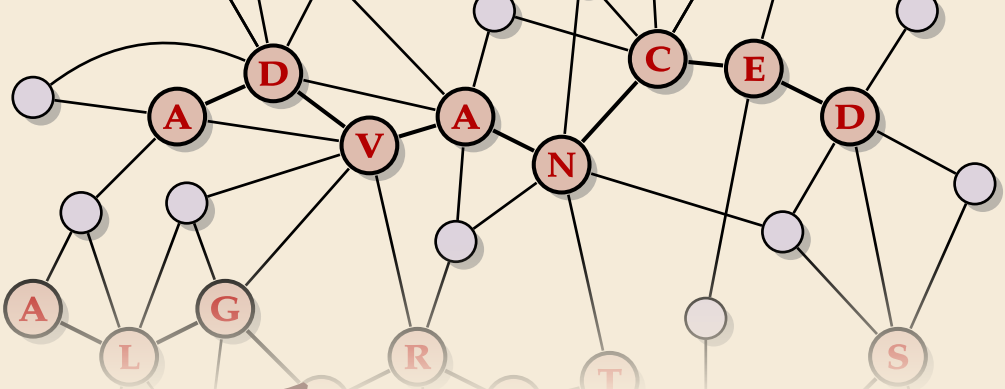


4 Fixed-Parameter Algorithms

Advanced Algorithms · Winter 2026–27

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Philosophy of FPT

- **Goal:** Principled theory for studying complexity based on two dimensions: input size $n = |x|$ (encoding length) and *some additional parameter* k
  - generalize ideas from $k = MaxInt(x)$
  - investigate influence of k (and n) on running time

  $\rightsquigarrow$ Try to find a parameter k such that
  (1) the problem can be solved efficiently as long as k is small, and
  (2) practical instances have small values of k (even where n gets big).

Motivation: Satisfiability

Consider Satisfiability of CNF formula *the drosophila melanogaster of complexity theory*

- general worst case: NP-complete
- k = #literals per clause
  - $k \leq 2 \quad \rightsquigarrow$ in P
  - $k \geq 3$ NP-complete
- k = #variables
  - $O(2^k \cdot n)$ time possible (try all assignments)
- k = #clauses?
- k = #literals?
- k = #ones in satisfying assignment
- k = structural property of formula
- for MAX-SAT, k = #optimal clauses to satisfy

Parameters

Definition 4.1 (Parameterization)

Let $\Pi = (L, Y, \Sigma)$ be a decision problem. A *parameterization* of Π is a polytime computable mapping $\kappa : L \to \mathbb{N}$. ◀

Definition 4.2 (Parameterized problem)

A *parameterized (decision) problem* is a pair (Π, κ) of a decision problem Π and a parameterization κ of Π. ◀

Examples

As before: Typically leave encoding implicit.

Definition 4.3 (p-variables-SAT)

Given: formula boolean ϕ (same as before)

Parameter: number of variables

Question: Is there a satisfying assignment $v : [n] \to \{0,1\}$? ◀

Definition 4.4 (p-Clique)

Given: graph $G = (V, E)$ and $k \in \mathbb{N}$

Parameter: k

Question: $\exists V' \subseteq V \;:\; |V'| \geq k \;\wedge\; \forall u, v \in V' : \{u, v\} \in E$? ◀

Definition 4.5 (p-Vertex-Cover)

Given: graph $G = (V, E)$ and $k \in \mathbb{N}$

Parameter: k

Question: $\exists S \subseteq V \;:\; |S| \leq k \;\wedge\; \forall e \in E : S \cap e \neq \emptyset$? ◀

Canonical Parameterization

Typical use case: study (decision versions of) optimization problems

Definition 4.6 (Canonical Parameterization)

If the instances of a decision problem Π are pairs (x, k) with $k \in \mathbb{N}$, the *canonical parameterization* is $\kappa(x, k) = k$. We write $p\text{-}\Pi$ for (Π, κ).

For $\Pi = (\Sigma_I, \Sigma_O, L, L_I, M, \mathit{value}, \mathit{goal})$ an optimization problem, $p\text{-}\Pi$ denotes the *(canonically) parameterized (decision) problem* given by the threshold problem $\mathit{Threshold}_\Pi$:
$p\text{-}\Pi = p\text{-}\mathit{Threshold}_\Pi$. ◀

Recall: $\mathit{Threshold}_\Pi$ is the set of pairs (x, k) of all instances $x \in L_I$ that have solutions that are weakly "better" than k.

Examples:

- p-Clique
- p-Vertex-Cover
- p-Graph-Coloring
- ...

Naming convention for other parameters:
p-*clause*-CNF-SAT: CNF-SAT with parameter "number of *clauses*"

4.1 Fixed-Parameter Tractability

Exemplary Running Times of Parameterized Problems

(consider simplest brute-force methods for problems)

- p-*variables*-SAT
  - k variables, n length of formula
  - $\rightsquigarrow$ $O(2^k \cdot n)$ running time
- p-Clique
  - k threshold (clique size); n vertices, m edges in graph
  - $\rightsquigarrow$ $\binom{n}{k}$ candidates to check, each takes time $O(k^2)$ to check
  - $\rightsquigarrow$ Total time $O(n^k \cdot k^2)$
- p-VertexCover
  - k threshold (VC size); n vertices, m edges in graph
  - $\rightsquigarrow$ $\binom{n}{k}$ candidates to check, each takes time $O(m)$ to check
  - $\rightsquigarrow$ Total time $O(n^k \cdot m)$
- p-GraphColoring
  - k threshold (#colors); n vertices, m edges in graph
  - $\rightsquigarrow$ k^n candidates to check, each takes time $O(m)$
  - $\rightsquigarrow$ Total time $O(k^n \cdot m)$

FPT Running Time

Definition 4.7 (fpt-algorithm)

Let (Π, κ) be a parameterized problem, $\Pi = (L, Y, \Sigma)$.
A (deterministic) algorithm A (with input alphabet Σ) is a *fixed-parameter tractable algorithm (fpt-algorithm)* w.r.t. κ if its running time on $x \in L$ with $\kappa(x) = k$ is at most

$$f(k) \cdot p(|x|) \quad = \quad O\big(f(k) \cdot |x|^c\big)$$

where p is a polynomial of degree c and f is an **arbitrary** computable function. ◀

Definition 4.8 (FPT)

A parameterized problem (Π, κ) is *fixed-parameter tractable* if there is an fpt-algorithm w.r.t. κ that decides Π.
The complexity class of all such problems is denoted by FPT. ◀

Intuitively, FPT plays the role of P.

A First FPT Example

Theorem 4.9 (p-variables-SAT is FPT)

p-variables-SAT $\in$ FPT. ◀

Proof:

Suffices to use brute force satisfiability for *p-variables*-SAT

```
1 procedure bruteForceSat(φ, X = {x_1, ..., x_k})
2     if k == 0
3         if φ == true return ∅ else UNSATISFIABLE
4     for value in {true, false} do
5         A := {x_1 ↦ value}
6         ψ := φ[x_1/value] // Substitute value for x_1
7         B := bruteForceSat(ψ, {x_2, ..., x_k})
8         if B ≠ UNSATISFIABLE
9             return A ∪ B
```

Worst case running time: $O(2^k\, n)$ for $n = |\varphi|$.
2^k recursive calls;
base case needs time $O(|\phi|)$ to check whether formula evaluates to *true* ■

. . . but #variables not usually small

Aren't we all FPT?

Theorem 4.10 (k never decreases → FPT)
Let $\Pi = (L, Y, \Sigma)$ be a decidable problem, $g : \mathbb{N} \to \mathbb{N}$ weakly increasing, unbounded and computable, and κ a parameterization of Π with

$$\forall x \in L : \kappa(x) \geq g(|x|).$$

Then $(\Pi, \kappa) \in \mathsf{FPT}$ (for *any* decidable Π!). ◀

g weakly increasing: $n \leq m \;\to\; g(n) \leq g(m)$
g unbounded: $\forall t \; \exists n \; : \; g(n) \geq t$

Proof:
Y decidable $\rightsquigarrow$ $\exists$ algorithm A deciding Π in time $\leq T(|x|)$ for *some* computable $T : \mathbb{N} \to \mathbb{N}$

w.l.o.g. T is weakly increasing and $T(n) \geq n$.

Idea: "hide" $T(|x|)$ inside $f(k)$: since $k = \kappa(x) \geq g(|x|)$ and g is unbounded, a large input forces a large parameter.

$\rightsquigarrow$ Define $h(k) \;:=\; \max\big(\{\, n \in \mathbb{N} : g(n) \leq k \,\} \cup \{1\}\big)$.

Aren't we all FPT? – Proof

Proof (cont.):

$h(k) \;:=\; \max\big(\{\, n \in \mathbb{N} : g(n) \le k \,\} \cup \{1\}\big).$

1. g weakly increasing and unbounded $\rightsquigarrow$ the set is finite $\rightsquigarrow$ h well-defined
2. h weakly increasing
3. g computable $\rightsquigarrow$ h computable (test $n = 1, 2, \ldots$ until $g(n) > k$)
4. $h(g(n)) \ge n$ for all n (as n is in the set for $k = g(n)$)

Now let x be an input, $n = |x|$, $k = \kappa(x) \ge g(n)$. Running A on x takes time

$$T(n) \;\le\; T\big(h(g(n))\big) \;\le\; T\big(h(k)\big) \;=:\; f(k),$$

using (4) and T increasing, then $g(n) \le k$ and h, T increasing.

f is computable, so A is an fpt-algorithm for (Π, κ) (with $p(n) \equiv 1$). ■

Back to “sensible” parameters

- ⇝ always check if parameter is reasonable (can be expected to be small)
  - if not, FPT might not even mean in NP!
- but now, for some positive examples!

4.2 Depth-Bounded Exhaustive Search I

FPT Design Pattern

- The simplest FPT algorithms use exhaustive search
- but with a search tree bounded by $f(k)$

- bruteforceSat was a typical example!
- does this work on other problems?

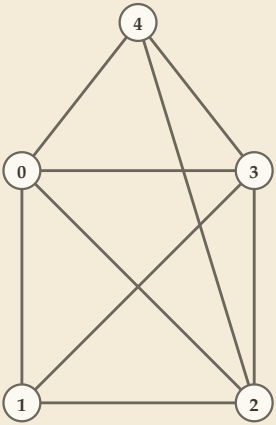
Depth-Bounded Search for Vertex Cover

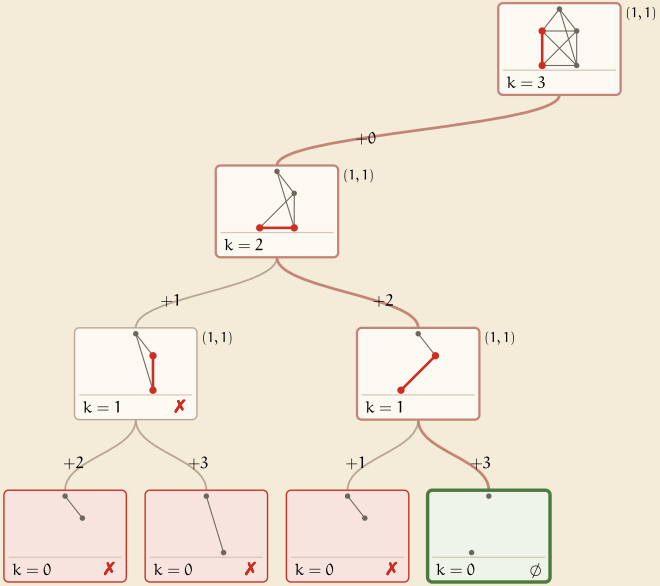
Let's try p-VertexCover.
Key insight: for every edge $\{v, w\}$, any vertex cover must contain v or w

```
1  procedure simpleFptVertexCover(G = (V, E), k):
2      if E == ∅ then return ∅
3      if k == 0 then return NOT_POSSIBLE  // truncate search
4      Choose {v, w} ∈ E (arbitrarily)
5      for u in {v, w} do:
6          G_u := (V \ {u}, E \ {{u, x} ∈ E})  // Remove u from G
7          C_u := simpleFptVertexCover(G_u, k − 1)
8      if C_v == NOT_POSSIBLE then return C_w ∪ {w}
9      if C_w == NOT_POSSIBLE then return C_v ∪ {v}
10     if |C_v| ≤ |C_w| then return C_v ∪ {v} else return C_w ∪ {w}
```

- Does not need explicit checks of solution candidates!
- runs in time $O\left(2^k(n+m)\right)$ $\rightsquigarrow$ fpt-algorithm for p-Vertex-Cover

Depth-Bounded Search for Vertex Cover – Example





Guessing the parameter

- Note: Previous algorithm only uses k to *truncate* branches.

$\rightsquigarrow$ We can *guess* a k and it still works

$\rightsquigarrow$ Try all k!

```
1 procedure vertexCoverBfs(G = (V, E))
2     for k := 0, 1, ..., |V| do
3         C := simpleFptVertexCover(G, k)
4         if C ≠ NOT_POSSIBLE return C
```

- Running time: $\sum_{k'=0}^{k} O(2^{k'}(n+m)) \;=\; O(2^k(n+m))$

$\rightsquigarrow$ For exponentially growing cost, trying all values up to k costs only constant factor more

4.3 Problem Kernels

Preprocessing

- Second key fpt technique are *reduction rules*
- **Idea:** Reduce the size of the instance (in polytime) without changing its outcome
- Trivial example for SAT:

  If a CNF formula contains a single-literal clause $\{x\}$ resp. $\{\neg x\}$, set x to *true* resp. *false* and remove the clause.

  - doesn't do anything in the worst case . . .
  - special case of resolution calculus rule $\dfrac{a_1 \vee a_2 \vee \cdots \vee x, \quad b_1 \vee b_2 \vee \cdots \vee \neg x}{a_1 \vee a_2 \vee \cdots \vee b_1 \vee b_2 \vee \cdots}$
  - basis of practical SAT solvers
- Trivial example for VertexCover

  include other endpoint in solution

  Remove vertices of degree 0 or 1. (never needed as part of optimal VC)
- Here: reduction rules that provably shrink an instance to size $g(k)$

Buss's Reduction Rule for VC

- Given a p-VertexCover instance (G, k)

  "deg > k" Rule: If G contains vertex v of degree $\deg(v) > k$, include v in potential solution and remove it from the graph.

- Can apply this simultaneously to degree $> k$ vertices.
- Either rule applies, or all vertices bounded degree(!)

Kernels

Definition 4.11 (Kernelization)

Let (Π, κ) be a parameterized problem, $\Pi = (L, Y, \Sigma)$.
A function $K : L \to L$ is a *kernelization* of (Π, κ) if it maps every instance $x \in L$ with $k = \kappa(x)$ to an instance $x' = K(x)$ so that

1. (self-reduction) $x \in Y \iff x' \in Y$
2. (polytime) K is computable in polytime.
3. (kernel-size) $|x'| \leq g(k)$ and $\kappa(x') \leq g(k)$ for some computable function g

We call x' the *(problem) kernel* of x and g the *size of the problem kernel*. ◀

Buss's Kernel

Buss's Reduction for Vertex Cover: (repeatedly apply until no more changes)

- $\deg > k$ rule
- Remove degree 0 and 1 vertices

Theorem 4.12 (Buss's Reduction is Kernelization)

Buss' reduction yields a kernelization for p-Vertex-Cover with kernel size $O(k^2)$. ◀

Proof:
After repeatedly applying Buss's rule as well as the isolated/leaf rule until neither applies further, we have $\forall v \in V : 2 \leq \deg(v) \leq k$.
(Note that the rule might reduce the parameter k).
In the resulting graph, any VC of size $\leq k$ covers $\leq k^2$ edges.
If $m > k^2$, we output a trivial No-instance (e. g., a K_{k+1} a complete graph on $k+1$ vertices).
If $m \leq k^2$, then the input size is now bounded by $g(k) = 2k^2$. ■

FPT iff Kernelization

Theorem 4.13 (FPT ↔ kernel)

Let (Π, κ) be a parameterized problem with Π decidable.
(Π, κ) is fixed-parameter tractable if and only if there is a kernelization of (Π, κ).

Proof:

FPT iff Kernelization [2]

Proof (cont.):

Max-SAT Kernel

Theorem 4.14 (Kernel for Max-SAT)

p-Max-SAT has a problem kernel of size $O(k^2)$ which can be constructed in linear time. ◀

Proof:

Max-SAT Kernel [2]

Proof (cont.):

Max-SAT Kernel [3]

Proof (cont.):

Corollary 4.15

p-Max-SAT $\in$ FPT

4.4 Depth-Bounded Search II: Planar Indep. Set

Deeper results (towards more shallow trees)

- Our previous examples of depth-bounded search were basically brute force
- Here we will see two more examples that exploit the problem structure in more interesting ways

- Recall: IndependentSet
- IndependentSet is NP-hard on planar graphs even with vertex degrees at most 3
- We will see: general problem p-Independent-Set likely not in FPT.

Independent Set on Planar Graphs

Definition 4.16 (p-Planar-Independent-Set)

Given: a *planar* graph $G = (V, E)$ and $k \in \mathbb{N}$
Parameter: k
Question: $\exists V' \subset V \quad : \quad |V'| \geq k \ \wedge \ \forall u, v \in V' : \{u, v\} \notin E$? ◀

Theorem 4.17 (Depth-Bounded Search for Planar Independent Set)

p-Planar-Independent-Set is in FPT and can be solved in time $O(6^k n)$. ◀

Elementary Knowledge on Planar Graphs

Theorem 4.18 (Euler's formula)

In any finite, *connected* (embedded) planar graph G with n vertices, m edges, and f faces holds $n - m + f = 2$. ◀

Corollary 4.19

A *simple* planar graph G on $n \geq 3$ nodes has $m \leq 3n - 6$ edges.
The average degree in G is < 6. ◀

Depth-Bounded Search for Planar Independent Set

```
1  procedure planarIndependentSet(G = (V, E), k):
2      if k == 0 then return ∅
3      if k > |V| then return NOT_POSSIBLE // truncate search
4      Choose v ∈ V with minimal degree; let w_1, ..., w_d be v's neighbors
5      // By planarity, we know d ≤ 5.
6      for u in {v, w_1, ..., w_d} do
7          D := {u} ∪ N(u)
8          G_u := (V \ D, E \ {{x, y} ∈ E : x ∈ D}) // Delete u and its neighbors
9          I'_u := planarIndependentSet(G_u, k − 1)
10         if I'_u ≠ NOT_POSSIBLE then return {u} ∪ I'_u
11     return NOT_POSSIBLE
```

Summary Planar Independent Set

- planarIndependentSet will often be faster than $O(6^k n)$
- works unchanged in $O((d+1)^k n)$ time for any degeneracy-d graph

every (induced) subgraph has vertex of degree at most d

4.5 Depth-Bounded Search III: Closest String

Closest String

Definition 4.20 (p-Closest-String)

Given: S set of m strings $s_1, s_2, \ldots, s_m$ of length L over alphabet Σ and a $k \in \mathbb{N}$.
Parameter: k
Question: Is there a string s for which $d_H(s, s_i) \leq k$ holds for all $i = 1, \ldots, m$? ◀

Dirty Columns

Definition 4.21 (Dirty Column)

A column of the $m \times L$ matrix corresponding to m strings of length L is called *dirty* if it contains at least 2 different symbols. ◀

Lemma 4.22 (Many Dirty Columns → No)

Let an instance to Closest-String with m strings of length L and parameter k be given.
If the corresponding $m \times L$ matrix contains more than $m \cdot k$ dirty columns, then no solution for the given instance exists. ◀

Depth-Bounded Search for Closest String

```
1  procedure closestStringFpt(s, d):
2      if d < 0 then return NOT_POSSIBLE
3      if d_H(s, s_i) > k + d for an i ∈ {1, ..., m} then
4          return NOT_POSSIBLE
5      if d_H(s, s_i) ≤ k for all i = 1, ..., m then return s
6      Choose i ∈ {1, ..., m} arbitrarily with d_H(s, s_i) > k
7          P := {p : s[p] ≠ s_i[p]}
8          Choose arbitrary P' ⊆ P with |P'| = k + 1
9          for p in P' do
10             s' := s
11             s'[p] := s_i[p]
12             s_ret := closestStringFpt(s', d − 1)
13             if s_ret ≠ NOT_POSSIBLE then return s_ret
14     return NOT_POSSIBLE
```

- initial call closestStringFpt(s_1, k)

Too Much Dirt

Lemma 4.23 (Pair Too Different → No)
Let $S = \{s_1, s_2, \ldots, s_m\}$ a set of strings and $k \in \mathbb{N}$. If there are $i, j \in \{1, \ldots, m\}$ with $d_H(s_i, s_j) > 2k$, then there is no string s with $\max_{1 \leq i \leq m} d_H(s, s_i) \leq k$. ◀

Depth-Bounded Search for Closest String

Theorem 4.24 (Search Tree for Closest String)

There is a search tree of size $O(k^k)$ for problem p-Closest-String. ◀

Corollary 4.25 (Closest String is FPT)

p-Closest-String can be solved in time $O(mL + mk \cdot k^k)$. ◀

- preprocessing ($O(mL)$ time)
  - ignore any clean columns
  - reject if more than mk dirty columns

⇝ effective string length after preprocessing is $L' \leq mk$

- call closestStringFpt(s_1, k)
  - maintain $d_H(s, s_i)$ in an array
    - ⇝ checking any distance $d_H(s, s_i)$ takes $O(1)$ time
    - before and after recursive call, update array to reflect $d_H(s', s_i)$
      Single character changed, so update only needs to check single position
    - ⇝ Can maintain distances in $O(m)$ time per recursive call
  - P' can be computed in $O(mk)$ time

4.6 Linear Recurrences & Better Vertex Cover

A Better Algorithm for Vertex Cover

Recall: Branching on endpoints of k edges gives search space of size 2^k for Vertex-Cover.
Can we do better?

Idea: Enlarge base case with "easy inputs"

Here: Consider graphs G with $\deg(v) \leq 2$ for all $v \in V(G)$.

Depth-Bounded Search for Vertex Cover

```
1  procedure betterFptVertexCover(G = (V, E), k):
2      if E = ∅ then return ∅
3      if k ≤ 0 then return NOT_POSSIBLE  // truncate search
4      if all nodes have degree ≤ 2 then
5          Find connected components of G
6          for each component G_i do
7              Fill C_i by picking every other node,
8              starting with the neighbor of a degree-one node if one exists
9          C := ⋃ C_i
10         if |C| ≤ k then return C else return NOT_POSSIBLE
11     Choose v with maximal degree, let w_1, ..., w_d be its neighbors // d ≥ 3
12     for D in {{v}, {w_1, ..., w_d}} with |D| ≤ k do:
13         G_D := G[V \ D] // Remove D from G
14         C'_D := betterFptVertexCover(G_D, k − |D|)
15         if C'_D ≠ NOT_POSSIBLE then C_D := D ∪ C'_D
16     return smallest C_D found, or NOT_POSSIBLE if none
```

How to analyze running time of betterFptVertexCover?

Analysis of betterFptVertexCover

worst case running time

- never have all degrees ≤ 2
- always need both recursive calls (until base case)
- ignore that graph gets smaller

$T_0 = \Theta(1)$
$T_k = \Theta(|V| + |E|) + T_{k-3} + T_{k-1}$

If we only number of base cases B_n, we obtain $T_n = O(B_n n^2)$

$B_0 = 1,\ B_1 = 1,\ B_2 = 1$
$B_k = B_{k-3} + B_{k-1} \qquad (k \geq 3)$

Solving Linear Recurrences

Example: $B_0 = B_1 = B_2 = 1, \qquad B_k = B_{k-1} + B_{k-3} \quad (k \geq 3)$

Tool: the *ordinary generating function* $B(z) := \sum_{k \geq 0} B_k z^k$

(a formal power series; $[z^k]\, B(z) = B_k$ extracts the coefficient)

Translate the recurrence into an equation for $B(z)$:

$$\begin{aligned} B(z) &= 1 + z + z^2 + \sum_{k \geq 3} \big(B_{k-1} + B_{k-3}\big) z^k \\ &= 1 + z + z^2 + z \sum_{k \geq 3} B_{k-1} z^{k-1} + z^3 \sum_{k \geq 3} B_{k-3} z^{k-3} \\ &= 1 + z + z^2 + z\big(B(z) - B_0 - B_1 z\big) + z^3 B(z) \\ &= 1 + z\, B(z) + z^3 B(z). \end{aligned}$$

$$B(z)\big(1 - z - z^3\big) = 1 \qquad \rightsquigarrow \qquad \boxed{B(z) = \frac{1}{1 - z - z^3}}$$

Solving Linear Recurrences [2]

Let z_0, z_1, z_2 be the roots of $1 - z - z^3$ (all distinct here): $z_0 \approx 0.68233$, $z_{1,2} \approx -0.34116 \pm 1.16154\,i$.

Partial fractions: $B(z) = \dfrac{1}{1-z-z^3} = \dfrac{A_0}{z-z_0} + \dfrac{A_1}{z-z_1} + \dfrac{A_2}{z-z_2}$ for constants A_0, A_1, A_2.

Each term is a geometric series:

$$\frac{A_j}{z-z_j} = \frac{-A_j/z_j}{1-z/z_j} = -\frac{A_j}{z_j}\sum_{k\ge 0}\left(\frac{1}{z_j}\right)^k z^k \qquad \rightsquigarrow \qquad [z^k]\,\frac{A_j}{z-z_j} = -\frac{A_j}{z_j}\left(\frac{1}{z_j}\right)^k.$$

$$B_k = [z^k]\,B(z) = -\frac{A_0}{z_0}\left(\frac{1}{z_0}\right)^k - \frac{A_1}{z_1}\left(\frac{1}{z_1}\right)^k - \frac{A_2}{z_2}\left(\frac{1}{z_2}\right)^k = \Theta\Big(\big(\tfrac{1}{z_0}\big)^k\Big),$$

since $|z_0| < |z_1| = |z_2|$ (the root closest to 0 dominates).

$\rightsquigarrow$ $B_k = \Theta\big(1.4656^k\big)$ with $1/z_0 \approx 1.4656$

Note: $1/z_0$ is the *largest* root of the *characteristic polynomial* $z^3 - z^2 - 1$ (substitute $z \mapsto 1/z$).

Solving Linear Recurrences – Result

Theorem 4.26 (Linear Recurrences)

Let $d_1, \ldots, d_i \in \mathbb{N}$ and $d = \max d_j$.
The solution to the *homogeneous linear recurrence equation*

$$T_n \quad = \quad T_{n-d_1} + T_{n-d_2} + \cdots + T_{n-d_i}, \qquad (n \geq d)$$

is always given by

$$T_n \quad = \quad \sum_{\ell} \sum_{j=0}^{\mu_\ell - 1} c_{\ell,j}\, z_\ell^n\, n^j$$

where we sum over all roots z_ℓ of multiplicity μ_ℓ of the so-called *characteristic polynomial* $z^d - z^{d-d_1} - z^{d-d_2} \cdots - z^{d-d_i}$.
The d coefficients $c_{\ell,j}$ are determined by the d initial values $T_0, T_1, \ldots, T_{d-1}$. ◀

Corollary 4.27

$T_n = O(z_0^n n^d)$ for z_0 the root of the characteristic polynomial with *largest absolute value*. ◀

Analysis of betterFptVertexCover [2]

$T_0 = \Theta(1)$,
$T_k = \Theta(n+m) + T_{k-3} + T_{k-1}$ $(n = |V|, m = |E|)$

Counting only leaves B_k of the search tree (internal nodes $\leq$ leaves): $T_k = O\big(B_k \cdot (n+m)\big)$

$B_0 = 1,\ B_1 = 1,\ B_2 = 1$
$B_k = B_{k-3} + B_{k-1}$ $(k \geq 3)$

$\rightsquigarrow$ $\vec{d} = (1, 3)$; characteristic polynomial $z^3 - z^2 - 1$
roots at $z_0 \approx 1.4656$ and $z_{1,2} \approx -0.2328 \pm 0.7926i$

Theorem 4.28 (Depth-Bounded Search for Vertex Cover)
p-Vertex-Cover can be solved in time $O(1.4656^k n^2)$. ◀

4.7 Interleaving

Motivation

Up to now, considered two-phase algorithms

1. Reduction to problem kernel
2. Solve kernel by depth-bounded exhaustive search

Idea: Apply kernelization *in each recursive step*.

(Extreme) Example: Vertex Cover with large-degree rule

- As a (slightly artificial) example, consider only using the simple reduction rule

  "deg > k" Rule: If G contains vertex v of degree $\deg(v) > k$, include v in potential solution and remove it from the graph.

- **Algorithm A:**
  1. Apply $\deg > k$ rule until saturation
  2. Call simpleFptVertexCover (recursively branch over arbitrary edge)
- **Algorithm B:** Same, interleaved:
  - Modified simpleFptVertexCover
  - Before choosing each new edge to branch on, apply $\deg > k$ rule.

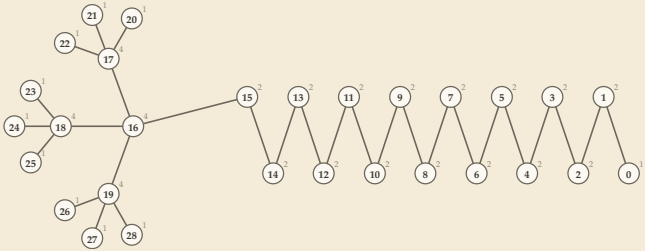
SimpleFptVertexCover Interleaved

```
1  procedure simpleFptVertexCover(G = (V, E), k):
2      if E == ∅ then return ∅
3      if k == 0 then return NOT_POSSIBLE
4      // nothing
5      // new
6      // on
7      // this
8      // side
9      Choose {v, w} ∈ E (arbitrarily)
10     for u in {v, w} do:
11         G_u := G[V \ {u}]
12         C_u := simpleFptVertexCover(G_u, k − 1)
13     if C_v == NOT_POSSIBLE then return C_w ∪ {w}
14     if C_w == NOT_POSSIBLE then return C_v ∪ {v}
15     if |C_v| ≤ |C_w| then
16         return C_v ∪ {v}
17     else
18         return C_w ∪ {w}
```

```
1  procedure simpleInterleavedVC(G = (V, E), k):
2      if E == ∅ then return ∅
3      if k == 0 then return NOT_POSSIBLE
4      C := ∅
5      while ∃v ∈ V : deg(v) > k
6          G := G[V \ {v}] // Remove v
7          C := C ∪ {v}
8          k := k − 1
9      Choose {v, w} ∈ E (arbitrarily)
10     for u in {v, w} do:
11         G_u := G[V \ {u}]
12         C_u := C ∪ simpleInterleavedVC(G_u, k − 1)
13     if C_v == NOT_POSSIBLE then return C_w ∪ {w}
14     if C_w == NOT_POSSIBLE then return C_v ∪ {v}
15     if |C_v| ≤ |C_w| then
16         return C_v ∪ {v}
17     else
18         return C_w ∪ {w}
```

Comparison on Lollipop Flowers

Consider family of graphs G_k "Lollipop Flowers":
"head" vertex with $k-2$ stars of $k-2$ leaves each attached + "tail" of $3k+1$ vertex path



$n = |V(G_k)| = (k-2)(k-1)+1 + 3k+1 = k^2+4$

Algorithm A

$\deg > k$ rule does nothing
search space remains 2^k
Answer No after exploring all branches

$\rightsquigarrow$ time $\Theta(2^k k^2)$

Algorithm B

initially same (no reduction)
after 2 edges removed from tail, parameter $k-2$
vertices in head have degree $k-1$
Output No (parameter 0, but tail edges left)

$\rightsquigarrow$ time $\Theta(k^2)$

Setting for Interleaving

Can we prove a general speedup?

Assumptions: (more restrictive than general kernelization!)

- K kernelization that
  - produces *kernel of size* $\leq q(k)$ for q a *polynomial*
  - in time $\leq p(n)$ for p a polynomial
- Branch in depth-bounded search tree
  - into i subproblems with branching vector $\vec{d} = (d_1, \ldots, d_i)$ (i. e., parameter in subproblems $k - d_1, \ldots, k - d_i$)
  - Branching is computed in time $\leq r(n)$ for r a polynomial

$\rightsquigarrow$ search space has size $O(\alpha^k)$.

$\rightsquigarrow$ Running time of two-phase approach on input x with $n = |x|$ and $k = \kappa(x)$:

$$O\Big(p(n) \;+\; r\big(q(k)\big) \cdot \alpha^k\Big)$$

With Interleaving

Generic interleaving:

1 **if** $|I| > c \cdot q(k)$ **then**
2 $\quad (I, k) := (I', k')$ where (I', k') forms a problem kernel *// Conditional Reduction*
3 **end**
4 replace (I, k) with $(I_1, k - d_1), (I_2, k - d_2), \ldots, (I_i, k - d_i)$ *// Branching*

$\rightsquigarrow$ Running time of interleaved approach on input x with $n = |x|$ and $k = \kappa(x)$ is at most T_k:

$$T_\ell \quad = \quad T_{\ell - d_1} + \cdots + T_{\ell - d_i} \; + \; p\big(q(\ell)\big) + r\big(q(\ell)\big)$$

Compare to non-interleaved version:

$$T_\ell \quad = \quad T_{\ell - d_1} + \cdots + T_{\ell - d_i} \; + \; r\big(q(k)\big)$$

Here the inhomogeneous term is constant w.r.t. ℓ, but depends on k
$\rightsquigarrow$ cannot ignore constant factors

Analysis of interleaved betterFptVertexCover [1]

Consider betterFptVertexCover from before, but with $\deg > k$ rule added.

- Initial call has unbounded n and m; after applying degree $0, 1, > k$ rules (in $O(n + m)$ time) size of graph $n + m = O(k^2)$
- interleaving $\rightsquigarrow$ graph also bounded recursively (in terms of new k)
- Recursive worst-case time after first reduction:
  $T_0 = \Theta(1)$
  $T_k = O(k^2) + T_{k-3} + T_{k-1}$

Inhomogenous Linear Recurrences

Example: $T_0 = T_1 = T_2 = 1, \qquad T_k = k^2 + T_{k-1} + T_{k-3} \quad (k \geq 3)$

Same translation as before, $T(z) \coloneqq \sum_{k\geq 0} T_k z^k$:

$$\begin{aligned} T(z) &= 1 + z + z^2 + \sum_{k\geq 3} \big(k^2 + T_{k-1} + T_{k-3}\big) z^k \\ &= 1 + z\big(T(z) - 1 - z\big) + z^3 T(z) + \sum_{k\geq 3} k^2 z^k. \end{aligned}$$

The inhomogeneous part is a *rational function with poles only at* $z = 1$:

$$\sum_{k\geq 0} z^k = \frac{1}{1-z}, \qquad \sum_{k\geq 1} k z^{k-1} = \frac{d}{dz}\frac{1}{1-z} = \frac{1}{(1-z)^2}, \qquad \sum_{k\geq 2} k(k-1) z^{k-2} = \frac{2}{(1-z)^3},$$

so $\sum_{k\geq 3} k^2 z^k = \dfrac{P(z)}{(1-z)^3}$ for a polynomial P.

Inhomogenous Linear Recurrences [2]

Solving for $T(z)$:

$$T(z) \;=\; \frac{Q(z)}{(1-z-z^3)\,(1-z)^3} \qquad \text{for a polynomial } Q.$$

$\rightsquigarrow$ Poles of $T(z)$: the roots z_0, z_1, z_2 of $1-z-z^3$ **and** $z=1$ (triple).

Dominant singularity is still $z_0 \approx 0.68 < 1$ $\rightsquigarrow$ $T_k = \Theta\big((1/z_0)^k\big) = \Theta(B_k)$.

General principle: an inhomogeneity f_k whose GF has no singularity inside $|z| \le |z_0|$ does not change the growth rate.

Inhomogenous Linear Recurrences Summary

Theorem 4.29 (Linear Recurrences II)

Let $d_1, \ldots, d_i \in \mathbb{N}$, $d = \max d_j$, and let $z_0 > 1$ be the real root of largest absolute value of $z^d - \sum_{j=1}^{i} z^{d-d_j}$. Consider the ***in****homogeneous linear recurrence*

$$T_n \quad = \quad T_{n-d_1} + T_{n-d_2} + \cdots + T_{n-d_i} + \boldsymbol{f_n}, \qquad (n \geq d)$$

with initial values $T_0, \ldots, T_{d-1} \in \mathbb{R}_{>0}$ and $(f_n)_{n\in\mathbb{N}}$ a sequence of positive numbers with $f_n = O\big((z_0 - \varepsilon)^n\big)$ for some fixed $\varepsilon > 0$ (e. g., any polynomial).
Then $T_n = O(T_n^0)$ where T_n^0 is defined as T_n with $f_n \equiv 0$. ◀

A Little Excursion: Singularity Analysis

General strategy: use generating functions for asymptotic approximations

Sequence Land		**Generating Function Land**
▶ number sequence $(a_n)_{n\geq 0}$	$\longrightarrow$	▶ (ordinary) generating function $A(z) = \sum_{n\geq 0} a_n z^n$
▶ recurrence equation	$\longrightarrow$	▶ (functional) equation for $A(z)$
		$\downarrow$ solve, simplify (e. g., partial fractions)
		$\rightsquigarrow$ closed form for $A(z)$
▶ closed form for a_n	$\longleftarrow$	▶ exact coefficients $[z^n]A(z)$
		OR approximate $A(z)$ near its *dominant singularity*
▶ *asymptotic* approximation $a_n = z_0^{-n} n^{\alpha-1}(1 \pm O(n^{-1}))$	$\longleftarrow$ transfer thms	$\rightsquigarrow$ *singular expansion* at $z = z_0$ $A(z) = f(z) \pm O((1 - z/z_0)^{-\alpha})$

O-Transfer

Theorem 4.30 (Transfer-Theorem of Singularity Analysis)

Assume $f(z)$ is Δ-analytic and admits the singular expansion

$$f(z) \;=\; g(z) \;\pm\; O\big((1-z)^{-\alpha}\big) \qquad (z \to 1)$$

with $\alpha \in \mathbb{R}$. Then

$$[z^n]\, f(z) \;=\; [z^n] g(z) \;\pm\; O\big(n^{\alpha-1}\big) \qquad (n \to \infty).$$

Possible Extensions

- (constant) coefficients $c_j \cdot T_{n-d_j}$ in recurrence
  $\rightsquigarrow$ different characteristic polynomial, same ideas
- *any* recurrence that leads to a representation of the generating function as a *singular expansion* around the dominant singularity.

$$\begin{aligned} f(z) &= c(1-z/z_0)^{-m} \pm O((1-z/z_0)^{-m+1}) \qquad (z \to z_0) \\ \rightsquigarrow [z^n]\, f(z) &= \frac{c}{(m-1)!} z_0^{-n} n^{m-1} \cdot \left(1 \pm O(n^{-1})\right) \qquad (n \to \infty) \end{aligned}$$

- other powers α in $1/(1-z)^\alpha$:

$$[z^n] \frac{1}{(1-\frac{z}{z_0})^\alpha} = \frac{z_0^{-n} n^{\alpha-1}}{\Gamma(\alpha)} \left(1 \pm O(n^{-1})\right) \qquad (n \to \infty) \qquad \begin{matrix} -\alpha \notin \mathbb{N}_0 \\ z_0 > 0 \end{matrix}$$

- much more! $\rightsquigarrow$ *analytic combinatorics*

Analysis of interleaved betterFptVertexCover [2]

- $T_0 = \Theta(1)$
  $T_k = O(k^2) + T_{k-3} + T_{k-1}$

$\rightsquigarrow$ $T_k = O(1.4656^k)$ (same characteristic polynomial)

- Total time: $O(1.4656^k + n + m)$

- The current record is $O^*(1.25284^k)$ time:

  D. G. Harris, N. S. Narayanaswamy: *A faster algorithm for Vertex Cover parameterized by solution size,* STACS **2024**

  before that, for more than a decade: $O(1.2738^k + kn)$

  J. Chen, I. A. Kanj, G. Xia: *Improved upper bounds for vertex cover,* Theoretical Computer Science 411(40–42), 3736–3756 **2010**

Summary

- Strategies for fpt algorithms
  - Use parameter to bound depth of exhaustive search
  - Use problem specific reduction rules to shrink input $\rightsquigarrow$ kernel(ization)s
- analysis of exact exponential searches often reduces to linear recurrences
  - generating functions!
- more clever branching reduces exponent of search space
- interleaving kernelization and exhaustive search improves polynomial parts